Native-Domain Reservoirs for Mixed Field Reuse

GameCult / Fensalir

Focused Technical Draft, May 2026

Abstract

Reservoir reuse in real-time rendering is usually applied after a visible surface or pixel sample has already been chosen. This paper studies a narrower but important case: mixed field renderers where the reusable sample is the visible field claim itself. In FENSALIR, tube fields, SDF surfaces, compact splat fields, and later density or sensor fields compete for presentation through one temporal/spatial reservoir. A texel-owned reservoir can store field id, depth, color, confidence, and RIS weights, but it cannot prove that a selected sample remains inside the target domain support after motion, occlusion, disocclusion, or resolution changes.

We propose a native-domain reservoir contract: every selected sample stores its producer-domain coordinate, support footprint, proposal measure, target value, and legal shift metadata. Pixels are consumers of this selected domain evidence, not owners of sample identity. The falsifiable claim is that this contract reduces temporal leakage and ghosting for mixed tube/SDF/splat fields compared with texel-owned reuse at similar sample budgets. We define the sample contract, target/proposal measures for heterogeneous field claims, reuse validation rules, required baselines, ablations, and visual diagnostics. This draft intentionally does not claim final results; it is the submission-shaped paper that the implementation must satisfy.

1 Introduction

ReSTIR and GRIS showed that a renderer can reuse a small number of representative samples by preserving their target and proposal probability story [1, 3]. Area ReSTIR sharpened the lesson for antialiasing and depth of field: when the integration domain is an area, the selected sample's actual subpixel and lens coordinates are part of the reservoir state [4]. A reservoir that forgets where inside the domain its selected sample lives is not allowed to reuse that sample as though support overlap were obvious.

FENSALIR applies that lesson to primary field evidence. The current engine contains a shared four-row GPU field reservoir for current-frame RIS, temporal reuse, spatial/domain reuse, and final resolve. TubeField and SDF contributors already share target/source-PDF proposal lanes, and an older nearest-depth-minus-coverage priority path has been cut. The live row ABI now stores selected sample coordinates, producer-domain coordinates, support footprint, domain kind, and shift kind alongside field id, travel, normal, guide confidence, proposal weights, and stats. That makes support overlap a first-class validation input rather than an inference from a texel.

This paper narrows the previous architecture manifesto into one testable claim:

> For mixed tube/SDF/splat field rendering, storing selected native-domain coordinates, support, and legal shift metadata in the reservoir reduces temporal leakage and ghosting under motion, occlusion, and resolution changes relative to texel-owned reuse with the same target/source-PDF RIS update.

The claim is deliberately small enough to kill. It can fail if the metadata cost outweighs the quality gain, if support validation rejects too much useful history, if target measures are incomparable across producers, or if a simpler TSR-style resolver plus producer-specific validation performs as well.

2 Scope

We evaluate only the core invariant. The broader FENSALIR architecture also includes semantic grammars, contribution trees, residency scheduling, sensor fusion, planetary flow, HDR presentation, and D3D12 resource discipline. Those systems motivate the domain model but are not the submission claim.

The first evaluated contributors are:

- **TubeField**: texture- or buffer-backed Catmull-Rom curve/tube fields with rolling-column identity, curve-local support, emission, and explicit motion when available;
- **SDF surface**: bounded analytic or proxy-evaluated signed distance surfaces with world/object-domain hit position, normal, material class, and travel;
- **compact splat field**: bounded anisotropic Form/Appearance splats with local support and projected footprint.

Density/extinction, sensor confidence, cube-sphere terrain, and learned residency are deferred. They are allowed to influence the data model, not the result claims.

3 Related Work

Reservoir reuse. ReSTIR DI and GI provide the practical reservoir shape for real-time candidate reuse [1, 2]. GRIS generalizes the estimator to varied domains and shift mappings [3]. RTXDI documents production pass boundaries for initial, temporal, and spatial reuse [6, 7].

Area and reconstruction. Area ReSTIR is closest in spirit because it treats subpixel and lens coordinates as sample identity for area-domain reuse [4, 5]. TSR and modern TAA systems emphasize rejection, disocclusion classification, sample-age accounting, and debug visualizations [8, 9]. Our distinction is that those diagnostics attach to selected field evidence before final presentation, not only to color history.

Field representations. Hart's sphere tracing gives the conservative-bound contract for implicit surfaces [10]. EWA splatting and 3D Gaussian Splatting show why anisotropic projected support is a useful primitive [11, 12, 13]. Geometry clipmaps and Nanite show the value of hierarchy cuts and resident summaries [14, 15]. We do not claim a new hierarchy algorithm here; we use these systems to constrain the support and baseline vocabulary.

4 Native-Domain Reservoir Contract

A texel-owned field reservoir stores a selected sample as though the pixel were the native domain. A native-domain reservoir instead stores a tuple

Contributor	Native coordinate u	Support Ω_s	Shift metadata
TubeField	subpixel position, source row/column, curve parameter, segment id	tube core radius, feather, projected curve support	rolling-column motion, curve replay/re-eval policy
SDF surface	object/world hit position, local object coordinate	SDF bound radius, normal cone, proxy extent	object motion, previous hit projection, visibility gate
Splat field	local splat center/sample coordinate, orientation frame	anisotropic envelope, projected footprint	domain lineage, local-frame transform, Jacobian if measure changes

Table 1: Minimum native-domain state for the first evaluated contributors.

$$r = (s, d, u, \Omega_s, \hat{p}(s), q(s), M, W, C, \sigma), \tag{1}$$

where s is the selected sample, d is its producer domain, u is the selected coordinate inside that domain, Ω_s is its support footprint, $\hat{p}$ is the target importance, q is the proposal density or finite proposal probability, M is represented candidate count, W is accumulated weight, C is final contribution weight, and σ stores legal shift and validation metadata.

4.1 Target Measures

The hard part is not the RIS update; it is defining comparable target measures for heterogeneous field claims. We use a projected expected-contribution target:

$$\hat{p}_i(c) = \int_{\Pi(\Omega_c)} \rho_i(x, c)\, e_i(x, c)\, v_i(x, c)\, q_i(x, c)\, dx, \tag{2}$$

where $\Pi(\Omega_c)$ is the projected support of candidate c, ρ_i is support/filter density for contributor i, e_i is estimated visible field or radiance change, v_i is visibility/occlusion validity, and q_i is producer confidence. This converts different producer domains into one unit: expected contribution to the resolved field over projected support.

For a finite proposal set P_i, the source probability is

$$q(c) = \Pr(c \mid c \in P_i), \tag{3}$$

usually $1/|P_i|$ for uniform finite structural proposals, with represented count preserving how many candidates the selected sample stands for. A producer may use $q = 1$ only for a singleton proposal measure whose represented count and target are already normalized to that singleton. This is the part of the machine that must stay boring and explicit. Otherwise different producer families launder incomparable units through the reservoir.

4.2 Update and Merge

The update is conventional:

Contributor	Practical target approximation	Proposal measure
TubeField	projected core coverage × confidence × linear luminance/emission relevance × visibility	finite structural proposal set from proxy/raster coverage or producer curve samples
SDF surface	projected covered area × depth/normal confidence × material/radiance relevance × visibility	one deterministic structural proposal per valid hit or finite proxy hit set
Splat field	projected anisotropic footprint × confidence × Form/Appearance contribution bound	stochastic or finite selected-cut proposal set over resident splats

Table 2: Initial target/proposal definitions. Continuous PDFs are used only when a producer actually samples from a continuous measure. Deterministic structural proposals use explicit finite-set proposal probabilities rather than a magic free sample.

$$w(c) = \frac{\hat{p}(c)}{q(c)}, \tag{4}$$

$$W \leftarrow W + w(c), \tag{5}$$

$$M \leftarrow M + m_c, \tag{6}$$

$$P(c \text{ selected}) = \frac{w(c)}{W}. \tag{7}$$

The contribution weight is

$$C = \frac{W}{M\hat{p}(s)}. \tag{8}$$

Merging another reservoir r_o is allowed only after shifting/re-evaluating its selected sample into the target domain. The merge probability is

$$P(r_o \text{ selected}) = \frac{W_o}{W + W_o}. \tag{9}$$

5 Reuse Validation

5.1 Texel-Owned Baseline

The baseline stores selected field id, travel/depth, normal, confidence, target/source-PDF lanes, and RIS stats per pixel row. Temporal reuse validates previous row-3 history by reprojected UV, field id, travel, normal, domain validity, and guide confidence. Spatial reuse samples neighboring rows and validates by field id, travel, normal, scalar support, and domain tag. This is stronger than naive TAA, but weaker than native-domain reuse because it cannot ask whether the selected sample coordinate still lies inside target support.

5.2 Native-Domain Reuse

Native-domain temporal reuse adds three checks:

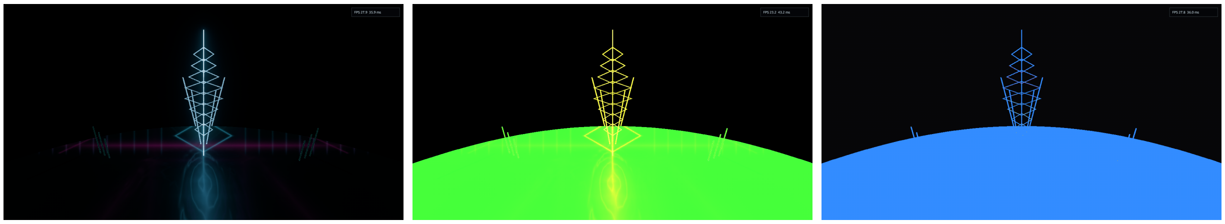

Figure 1: Prototype debug captures from the current shared field reservoir: final weighted resolve, proposal debug, and rejection codes. The key observation is not that these prove the method; they expose why support overlap and selected producer-domain coordinates must become visible state.

1. **Support overlap.** The previous selected coordinate u_p is transformed into the current domain. Reuse requires $T_{p \to c}(u_p) \in \Omega_c$, or a legal shift that creates a current-domain coordinate.
2. **Producer re-evaluation.** TubeField and SDF samples are re-evaluated at the shifted coordinate. A stored depth or color is not enough authority; the producer-domain sample must still produce a compatible claim.
3. **Measure correction.** If the shift changes measure, the sample carries a Jacobian or an explicit finite-set/MIS correction. If no correction is defined, the shift is invalid.

Spatial reuse follows the same rule with neighbor domains: sample first, validate domain mapping second, merge third. A neighbor row is never a visible fallback by itself.

6 Prototype Evidence

Figure 1 shows current Fensalir debug captures from the row-stage reservoir rebuild. These captures are not the final evaluation: they show the current diagnostic surface and the failure class that motivates native-domain state. The final paper must replace this figure with controlled side-by-side comparisons and ablations.

The first native-domain ABI cut expands each live `FieldReservoirSample` row from seven to nine `float4` lanes:

Lane	Meaning
`colorTravel`	resolved HDR contribution and camera travel
`metadata`	field id and normal/gradient payload
`control`	coverage, scalar support, detail, age/value payload
`guide`	confidence, sample age, domain validity, rejection code
`motion`	previous UV, expected previous travel, explicit-motion flag
`domainSample`	selected pixel/sample UV and packed producer coordinate
`domainSupport`	support footprint, domain kind, shift kind
`proposal`	target, source PDF, represented count, proposal kind
`stats`	selected target, weight sum, candidate count, contribution weight

TubeField row-0 proposals write the selected subpixel UV, logical source column, curve sample coordinate, tube-core support footprint, and explicit motion shift kind when rolling-column replay is available. The field id still uses physical rolling-column identity so age-column history cannot silently cross-validate. SDF scene proposals reconstruct the current world hit, pack object-local coordinates, and mark the sample as object-replayable. Reservoir validity now fails when a live sample lacks domain state, and temporal/spatial validation multiplies the existing field/travel/normal/confidence checks by support overlap in the selected-domain lane. The first TubeField replay path re-evaluates the selected logical column/curve coordinate against the live producer buffer through a bounded producer-keyed replay manifest and raw source descriptor table; selected samples choose the matching replay entry by field-id family instead of relying on a CPU-side single-batch binding. Manifest overflow is invalid rather than guessed. SDF temporal reuse reconstructs the previous selected world hit from stored sample UV/travel, carries the hit through object center motion, and rejects the history unless the replayed hit still matches the current camera ray, travel, and conservative object support. Exact client SDF distance re-evaluation remains producer-owned work rather than a generic postprocess authority. Debug modes expose selected coordinates, support footprint/domain kind, and shift/rejection state from the same row-3 layer consumed by presentation. The reservoir is also now a budgeted work grid rather than a native-resolution buffer: `GraphicsSettings.FieldReservoirScale` clamps scene evidence, candidate/history rows, reservoir resolve, bloom sources, and match-window graph targets below final presentation resolution by default. The full-resolution surface is reserved for presentation reconstruction and diagnostics. The renderer also exposes a field-reservoir mode switch for comparison: native mode enables domain validity, support overlap, and producer replay during temporal reuse, while texel-baseline mode keeps the same rows but disables those native-domain gates. The first paired capture set `reservoir-compare-20260529-231904-*` changed 10.8208% of final-frame pixels between modes, and `scripts/measure-reservoir-mode-comparison.ps1` now repeats that measurement. This proves the comparison switch is live; it is not yet a temporal leakage or ghosting metric.

7 Required Evaluation

This paper is not submission-ready until the following evaluation exists. Reviewers should reject the claim without it. That sentence stays here until the figures and tables replace it.

7.1 Scenes

1. **Occluded spectrum tubes**: dense rolling TubeField columns passing behind/through an SDF occluder.
2. **Mixed SDF and splat surface**: an analytic SDF object with compact splat detail and camera/subpixel motion.
3. **Resolution sweep**: identical field content at multiple output resolutions with fixed world-space support.
4. **Disocclusion stress**: fast camera/object motion exposing and hiding curve/SDF intersections.
5. **Near-equal-depth conflict**: overlapping field claims at similar travel to test RIS selection versus structural occlusion gates.

7.2 Baselines

- raw current-frame rendering without temporal reuse;
- conventional TAA/TSR-style texel history using depth/normal/field-id validation;

- texel-owned field reservoir with target/source-PDF lanes but no selected native coordinate;
- producer-specific TubeField reuse path;
- native-domain reservoir with support/shift metadata.

7.3 Ablations

- no selected domain coordinate;
- selected coordinate but no support overlap test;
- support overlap but no producer re-evaluation;
- no measure/Jacobian/MIS correction for shifted samples;
- no confidence/variance feedback to reconstruction;
- no current-frame structural occlusion gate before RIS.

7.4 Metrics

The evaluation must report:

- temporal leakage pixels in occluded/background regions;
- ghosting duration after disocclusion;
- support-miss rejection precision/recall against ground-truth producer evaluation;
- resolved image error against high-sample offline or high-budget reference;
- stochastic speckle variance over stable regions;
- GPU time, memory per reservoir row, bandwidth, and candidate counts;
- failure cases where native-domain validation rejects useful history.

8 Novelty Boundary

The estimator itself is not novel. The RIS update is standard. The idea of shift mappings is not novel after GRIS. Area support is not novel after Area ReSTIR. Hierarchical fields are not novel after clipmaps, Nanite, and splat hierarchies.

The claimed contribution is their boundary in a mixed field renderer: primary-visible field claims from heterogeneous producers share one reservoir only if the selected sample stores producer-domain coordinate, support, and legal shift metadata. The reservoir is earlier than a lighting/path reservoir and deeper than a texel history cache. That is the thing the evaluation must prove useful.

9 Limitations

The contract increases reservoir state. It may be too expensive for simple opaque scenes where conventional velocity/depth history already works. Target normalization across heterogeneous producers is an open risk; bad targets can make the reservoir mathematically clean and visually wrong. Finite structural proposal measures must be specified carefully, especially when proxy rasterization creates deterministic candidates. Finally, native-domain validation can increase shimmer if it rejects history more aggressively than the reconstruction pass can repair.

10 Conclusion

The paper lives or dies on one invariant: a reusable field reservoir sample must remember where it was selected in its producer domain and what support it represents. Without that, temporal/spatial reuse is a texel-owned cache with better vocabulary. With it, mixed tube/SDF/splat fields can be validated, shifted, rejected, reconstructed, and debugged at the layer where the visible claim actually lives. The next implementation pass must build the missing lane and produce the baselines, ablations, and images listed above. Until then, this is a sharp claim with unpaid proof obligations, not a finished paper.

References

[1] B. Bitterli, C. Wyman, M. Pharr, P. Shirley, A. Lefohn, and W. Jarosz. Spatiotemporal reservoir resampling for real-time ray tracing with dynamic direct lighting. *ACM Transactions on Graphics*, 2020. https://research.nvidia.com/labs/rtr/publication/bitterli2020spatiotemporal/

[2] Y. Ouyang, S. Liu, M. Kettunen, M. Pharr, and J. Pantaleoni. ReSTIR GI: Path resampling for real-time path tracing. *Computer Graphics Forum*, 2021. https://research.nvidia.com/publication/2021-06_restir-gi-path-resampling-real-time-path-tracing

[3] D. Lin, M. Kettunen, B. Bitterli, J. Pantaleoni, C. Yuksel, and C. Wyman. Generalized resampled importance sampling: Foundations of ReSTIR. *ACM Transactions on Graphics*, 2022. https://graphics.cs.utah.edu/research/projects/gris/sig22_GRIS.pdf

[4] S. Zhang, D. Lin, M. Kettunen, C. Yuksel, and C. Wyman. Area ReSTIR: Resampling for real-time defocus and antialiasing. *ACM Transactions on Graphics*, 2024. https://graphics.cs.utah.edu/research/projects/area-restir/AreaReSTIR-SIGGRAPH2024.pdf

[5] S. Zhang. Area-ReSTIR source code repository. https://github.com/guiqi134/Area-ReSTIR

[6] NVIDIA. RTXDI ReSTIR GI documentation. https://github.com/NVIDIA-RTX/RTXDI/blob/main/Doc/RestirGI.md

[7] NVIDIA. RTXDI ReSTIR PT documentation. https://github.com/NVIDIA-RTX/RTXDI/blob/main/Doc/RestirPT.md

[8] Epic Games. Temporal Super Resolution in Unreal Engine. https://dev.epicgames.com/documentation/unreal-engine/temporal-super-resolution-in-unreal-engine

[9] Epic Games. Temporal Super Resolution Frequently Asked Questions. https://dev.epicgames.com/documentation/unreal-engine/temporal-super-resolution-frequently-asked-questions-for-unreal-engine

[10] J. C. Hart. Sphere tracing: A geometric method for the antialiased ray tracing of implicit surfaces. *The Visual Computer*, 1996. https://experts.illinois.edu/en/publications/sphere-tracing-a-geometric-method-for-the-antialiased-ray-tracing/

[11] M. Zwicker, H. Pfister, J. van Baar, and M. Gross. EWA splatting. *IEEE Visualization*, 2001. https://dash.harvard.edu/bitstream/handle/1/4138240/Zwicker_EWA.pdf

[12] L. Ren, H. Pfister, and M. Zwicker. Object space EWA surface splatting. *Computer Graphics Forum*, 2002. CMU publication page

[13] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 2023. https://arxiv.org/abs/2308.04079

[14] A. Asirvatham and H. Hoppe. Terrain rendering using GPU-based geometry clipmaps. *GPU Gems 2*, 2005. https://developer.nvidia.com/gpugems/gpugems2/part-i-geometric-complexity/chapter-2-terrain-rendering-using-gpu-based-geometry

[15] B. Karis, R. Stubbe, and G. Wihlidal et al. A deep dive into Nanite virtualized geometry. *Advances in Real-Time Rendering*, SIGGRAPH 2021. https://advances.realtimerendering.com/s2021/Karis_Nanite_SIGGRAPH_Advances_2021_final.pdf